

SUMMER INTERNSHIP REPORT

Full Stack Web Development | CodeAlpha

Kartikey Gupta | 25SCS1003004592 | 2CSE37

10 July 2026 - 10 August 2026

1. Candidate's Declaration

I, Kartikey Gupta, declare that this report on ConnectHub Social Media Platform is my authentic work completed during the CodeAlpha Full Stack Development Virtual Internship (10 July - 10 August 2026).

2. Acknowledgement

I thank CodeAlpha mentors and support team for guidance during the internship and my institute faculty for their encouragement.

3. Internship Completion Certificate

Certificate issued on 1 September 2026 by CodeAlpha for Full Stack Development. Student ID: CA/DF1/194583. Letter of recommendation included in repository.

4.1 Introduction

During the internship I built ConnectHub, a full-stack social media web application using Node.js, Express, MongoDB, and vanilla JavaScript.

4.2 Organization Profile

CodeAlpha offers virtual internships in software development. Website: www.codealpha.tech | Email: services@codealpha.tech

4.3 Problem Statement

Users need a platform to register, create posts, interact through likes and comments, follow other users, and view a personalized feed.

4.5 Scope

Includes auth, posts, comments, likes, profiles, follow/unfollow, and personalized feed. Admin panel and real-time chat were out of scope.

4.6 Technologies Used

HTML5, CSS3, JavaScript, Node.js, Express, MongoDB, Mongoose, JWT, bcryptjs, helmet, cors, express-rate-limit, express-validator.

4.7 System Architecture

Three-tier: client (HTML/CSS/JS), Express REST API with middleware, MongoDB for Users, Posts, Comments. JWT stored in httpOnly cookies.

4.8 Methodology

Week 1: setup and auth. Week 2: posts and feed. Week 3: likes, comments, profiles. Week 4: follow system, testing, documentation.

4.10 Implementation Details

Auth: register, login, logout, /api/auth/me. Users: profile, follow/unfollow, followers/following. Posts: CRUD, like/unlike, feed. Comments: add/list/delete.

6. Conclusion

The internship strengthened my full stack skills in API design, authentication, database modeling, and building interactive web applications.

4.4 Project Objectives

- User registration and JWT authentication
- Create/view/delete posts
- Personalized home feed
- Like/unlike and comment on posts
- User profiles and follow system

5. Learning Outcomes

- REST API design
- MongoDB schema relationships
- Secure JWT auth with httpOnly cookies
- Frontend-backend integration
- Project documentation on GitHub

7. References

- Node.js Docs
- Express.js Guide
- MongoDB Manual
- MDN Web Docs
- CodeAlpha - www.codealpha.tech